MTH 451/551, Fall 2026
Quiz #2, Tue., Oct. 20, 2026

Solutions

1. Let $f(x) = x^3 - 2$ on the interval $[a, b] = [1, 2]$. You may find the following values of f useful:

x	1	1.25	1.5	1.75	2
$f(x)$	-1	-0.046875	1.375	3.359375	6

(a) Perform the bisection method to find x_0, x_1, and x_2, giving the interval $[a_n, b_n]$ at each step.

(b) How many iterations n are needed to guarantee that $|x_n - r| \leq 2^{-20} = (1{,}048{,}576)^{-1} \approx 10^{-6}$, where r is the root of f?

Solution. From the table, $f(1) < 0$ and $f(2) > 0$, so $[a_0, b_0] = [1, 2]$ and $x_0 = 1.5$. Since $f(1.5) > 0$, $f(a_0)f(x_0) \leq 0$, so $[a_1, b_1] = [1, 1.5]$ and $x_1 = 1.25$. Since $f(1.25) < 0$, $f(a_1)f(x_1) > 0$, so $[a_2, b_2] = [1.25, 1.5]$ and $x_2 = 1.375$.

By the convergence theorem, $|x_n - r| \leq 2^{-(n+1)}(b - a) = 2^{-(n+1)}$. This is at most 2^{-20} when $n + 1 \geq 20$, so $n = 19$ iterations suffice.

2. Now suppose we run the bisection algorithm from Problem 1 on a computer, using double-precision floating point arithmetic. We choose to terminate the algorithm only when $b_n - a_n \leq \text{xtol}$.

(a) Note $r = \sqrt[3]{2}$ is irrational. Is $r \in \mathbb{F}$? Give an upper bound on the minimum achievable error $|x_n - r|$.

(b) What happens to the algorithm when, for example, $\text{xtol} = 2^{-60}$?

Solution. (a) Every number in $\mathbb{F}$ has a finite binary expansion, and so is rational. Since $r = \sqrt[3]{2}$ is irrational, $r \notin \mathbb{F}$. Every iterate x_n is in $\mathbb{F}$, so $|x_n - r| > 0$ for all n, no matter how many iterations we perform. The floating point numbers in $[1, 2]$ are evenly spaced, with spacing $\epsilon = 2^{-52}$, and r lies between two consecutive ones. The closer of the two is $\text{fl}(r)$, which satisfies

$$0 < |\,\text{fl}(r) - r| \leq \tfrac{1}{2}\epsilon = 2^{-53} = u.$$

So the smallest achievable error is positive, and at most $u \approx 1.1 \times 10^{-16}$.

(b) In exact arithmetic, $b_n - a_n = 2^{-n}$, so we would terminate once $2^{-n} \leq \text{xtol}$, i.e. after $n = 60$ iterations when $\text{xtol} = 2^{-60}$. In floating point arithmetic, after $n = 52$ iterations, $b_{52} - a_{52} = 2^{-52} = \epsilon$, so a_{52} and b_{52} are consecutive floating point numbers, with no element of $\mathbb{F}$ strictly between them. The computed midpoint $\text{fl}((a_{52} + b_{52})/2)$ is therefore equal to either a_{52} or b_{52}, and in either case the bisection update leaves the interval unchanged. So $b_n - a_n = 2^{-52} > \text{xtol}$ for all $n \geq 52$, and the algorithm never terminates (or runs until maxit). The same happens for any $\text{xtol} < 2^{-52}$: the convergence theorem holds in exact arithmetic, but in floating point arithmetic the interval cannot shrink below the spacing of $\mathbb{F}$ near the root.